

Auteur: Abdellah El Moussaoui
Studierichting: Informatica
Oefeningenreeks 1C nr. 46.12

Trek de vierdemachtswortels in $\mathbb{C}$:

$$z^4 = 16 \operatorname{cis} \left(\frac{4\pi}{9} \right)$$

$$\Rightarrow z_k = 2 \operatorname{cis} \left(\frac{\frac{4\pi}{9} + 2k\pi}{4} \right), k = 0, 1, 2, 3$$

$$\Rightarrow z_k = 2 \operatorname{cis} \left(\frac{\pi}{9} + \frac{k\pi}{2} \right), k = 0, 1, 2, 3$$

$$\Rightarrow z_0 = 2 \operatorname{cis} \left(\frac{\pi}{9} \right)$$

$$\Rightarrow z_1 = 2 \operatorname{cis} \left(\frac{11\pi}{18} \right)$$

$$\Rightarrow z_2 = 2 \operatorname{cis} \left(\frac{10\pi}{9} \right)$$

$$\Rightarrow z_3 = 2 \operatorname{cis} \left(\frac{29\pi}{18} \right)$$

$$\Rightarrow \boxed{z_0 = 2 \operatorname{cis} \left(\frac{\pi}{9} \right), z_1 = 2 \operatorname{cis} \left(\frac{11\pi}{18} \right), z_2 = 2 \operatorname{cis} \left(\frac{10\pi}{9} \right), z_3 = 2 \operatorname{cis} \left(\frac{29\pi}{18} \right)}$$

References